Channel Independence Wins All? Reproducing and Extending PatchTST

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1 Introduction

Two recent transformer-based long-term time series forecasting (LTSF) methods offer opposing inductive biases. PatchTST [1] segments each univariate channel into patches and applies a shared Transformer encoder independently per channel (*channel independence*, CI), while iTransformer [2] treats each variate's full history as a single token and attends across channels (*channel dependence*, CD). Both papers claim state-of-the-art results on the same multivariate benchmarks, yet their conclusions directly contradict each other.

We reproduce PatchTST's core results and conduct a head-to-head comparison with iTransformer under a matched 20-epoch compute budget and unified training harness. We further include DLinear [3], a parameter-efficient (non-attention based) decomposition baseline, and propose **Stochastic Channel Mixing (SCM)**, a lightweight module that interpolates between CI and CD to explore whether controlled partial mixing yields additional gains.

2 Chosen Result

We target **Table 1** of Nie et al. [1], which reports MSE and MAE for PatchTST on Weather, Electricity, and Traffic at forecast horizons {96, 192, 336, 720}. This table is the paper's headline claim—that patch-based CI transformers outperform prior art—and iTransformer's own Table 1 directly contests it, making a controlled re-evaluation both natural and necessary.

3 Methodology

Model architectural design can be visualized as in Fig. 1.

PatchTST: Input series are segmented into patches (length 16, stride 8). Each patch serves as a token processed by a Transformer encoder with multi-head self-attention and feed-forward layers, applied independently per channel. A linear head maps final representations to the forecast horizon.

iTransformer: The input is transposed so that each variate's full look-back window forms one token. Self-attention operates across variates (CD), while feed-forward layers model temporal patterns per channel.

DLinear: A moving-average kernel decomposes the series into trend and seasonal components, each mapped by a separate linear layer. Despite minimal parameters, DLinear serves as a strong no-attention baseline.

Stochastic Channel Mixing (SCM): On top of PatchTST's CI encoding $\hat{e}_{\mathrm{CI}}$, a lightweight linear mixer produces a cross-channel blended encoding $\hat{e}_{\mathrm{CM}}$. The final encoding used in the same transformer model(PatchTST42) interpolates between the two:

$$\hat{e} = (1-\alpha)\,\hat{e}_{\mathrm{CI}} + \alpha\,\hat{e}_{\mathrm{CM}}, \quad \alpha \in \{0,\ 0.25,\ 0.50,\ 0.75,\ 1.0\}. \tag{1}$$

Unified evaluation: All models share identical data loaders, reversible instance normalization, and evaluation routines. In comparison to the paper [1], we use gradient accumulation and reduced batchsize to maintain the same effective batchsize ($batch\_size * accumulation\_steps$) as that in the paper. Training is capped at 20 epochs with matched hyperparameter grids across three datasets: Weather (M=21), Electricity (M=321), and Traffic (M=862).

4 Results & Analysis

4.1 Reproduction Quality and Main Comparison

Table 1 reports MSE across all 12 dataset–horizon cells. PatchTST achieves the lowest MSE on **all 12 cells**; iTransformer never wins. Our reproduced PatchTST values fall within 1–2% of the original paper on Weather and Electricity, with a slightly larger gap on Traffic(large dataset which needs more training), consistent with the reduced 20-epoch budget. DLinear remains competitive at shorter horizons (96, 192), reinforcing that simple baselines should not be overlooked.

Although PatchTST leads across the board, margins on Traffic are narrow (e.g., 0.388 vs. 0.393 at horizon 96), and these are single-seed results—multi-seed evaluation would be needed to confirm statistical significance.

4.2 When Does CI Work?

Cross-variate correlation increases with dataset dimensionality (Weather: 0.29, Electricity: 0.41, Traffic: 0.56). As shown in Figure 2 and Figure 3, CI's advantage narrows as correlation grows—from ∼12% MSE reduction on Weather to ∼2% on Traffic—but never disappears within this benchmark set. This suggests CI is a robust default for low-to-moderate correlation regimes, while high-correlation settings may benefit from controlled mixing.

4.3 Stochastic Channel Mixing

The SCM sweep (Figure 4) reveals a U-shaped performance curve. Light mixing at α=0.25 yields a 0.6% MSE reduction over pure CI, consistent across horizons. At α=0.50 gains vanish, and $\alpha \geq 0.75$ degrades performance below the CI baseline. This confirms that CI is near-optimal but not strictly so: a small dose of cross-channel information helps, but heavy mixing hurts generalization—likely because it reintroduces the overfitting risk that CI was designed to avoid.

5 Reflections

Standardized evaluation resolves contradictions: PatchTST and iTransformer each claim superiority on the same benchmarks, yet use different training schedules, normalization schemes, and data splits. A shared evaluation framework is essential for meaningful comparison.

Relative rankings converge before absolute performance: Our 20-epoch budget raised absolute MSE by roughly 2-8% relative to the original 100-epoch results, yet the CI > CD ordering held across all cells. This supports compute-matched comparison as a practical evaluation paradigm when full-budget runs are infeasible.

CI vs. CD is a continuum, not a binary choice: The SCM results show that treating mixing intensity as a continuous hyperparameter is more productive than committing to one inductive bias. The optimal α appears dataset-dependent and correlates with inter-channel correlation, pointing towards data-driven architecture selection.

Future directions: Extending to 100 epochs with multi-seed runs would test whether CI's dominance holds at convergence. A finer-grained α sweep (0.1 increments) across all datasets would map dataset-specific blending profiles. Most ambitiously, replacing fixed α with a learnable function of input statistics (e.g., inter-channel correlation) could fully automate the CI/CD trade-off.

References

[1] Y. Nie, N. H. A. Nguyen, P. Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. *ICLR*, 2023.

[2] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long. iTransformer: Inverted transformers are effective for time series forecasting. *ICLR*, 2024.

[3] A. Zeng, M. Chen, L. Zhang, and Q. Xu. Are transformers effective for time series forecasting? *AAAI*, 37(9), 2023.

[4] H. Wu, J. Xu, J. Wang, and M. Long. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *NeurIPS*, 2021.

Appendix

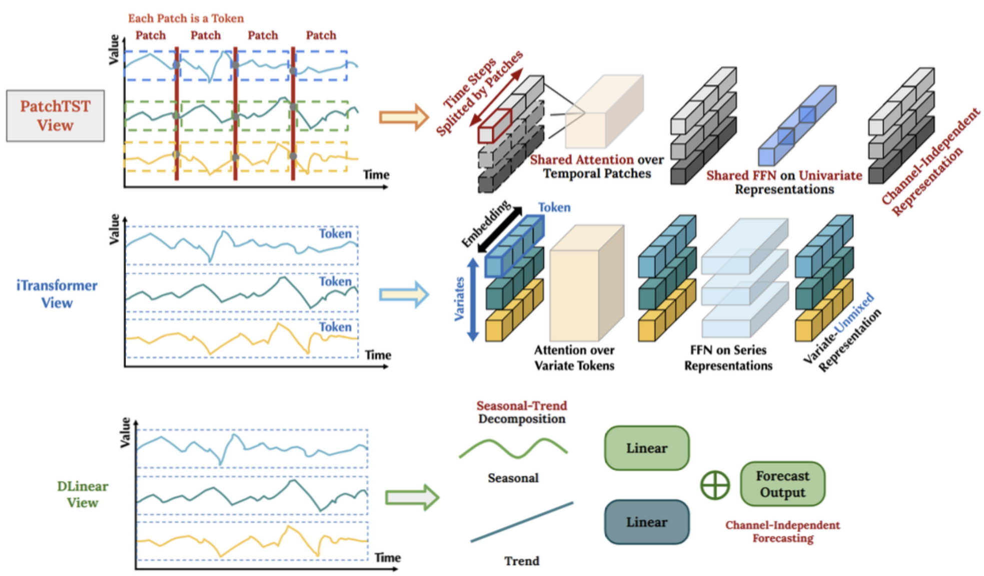

Figure 1: Model Architecture

Dataset	**M**	**H**	**PatchTST/64 [Paper]**	**PatchTST/42 [Paper]**	**PatchTST/64 [Ours]**	**PatchTST/42 [Ours]**	**iTransformer [Paper]**	**DLinear-I (CI)**	**DLinear (CD)**
Weather	21	96	0.149	0.152	0.146	0.155	0.176	0.156	0.1800
		192	0.194	0.197	0.189	0.196	0.225	0.200	0.2203
		336	0.245	0.249	0.242	0.250	0.281	0.251	0.2646
		720	0.314	0.320	0.317	0.318	0.358	0.326	0.3293
Electricity	321	96	0.129	0.130	0.152	0.133	0.148	0.140	0.1441
		192	0.147	0.148	0.167	0.149	0.166	0.154	0.1558
		336	0.163	0.167	0.180	0.166	0.179	0.167	0.1684
		720	0.197	0.202	0.224	0.204	0.209	0.212	0.2052
Traffic	862	96	0.360	0.367	0.396	0.388	0.393	0.5043	0.4320
		192	0.379	0.385	0.412	0.404	0.412	0.5195	0.4513
		336	0.392	0.398	0.416	0.415	0.425	0.5350	0.4652
		720	0.432	0.434	0.447	0.444	0.459	0.6042	0.4889

Table 1: MSE comparison under matched 20-epoch compute.

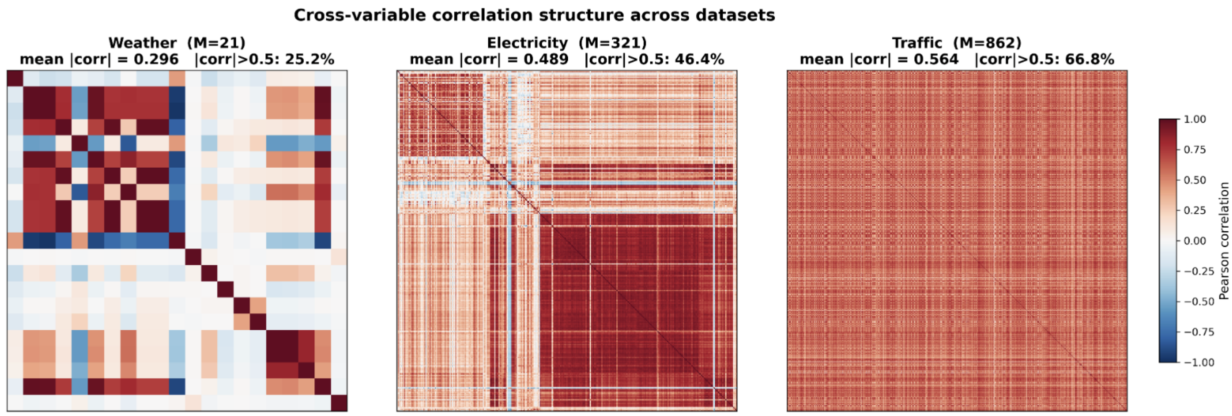

Figure 2: Cross-variate correlation structure across datasets

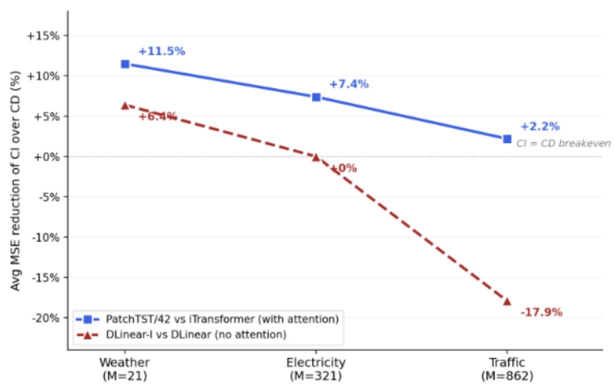

Figure 3: Relative MSE reduction of CI over CD narrows as mean correlation increases.

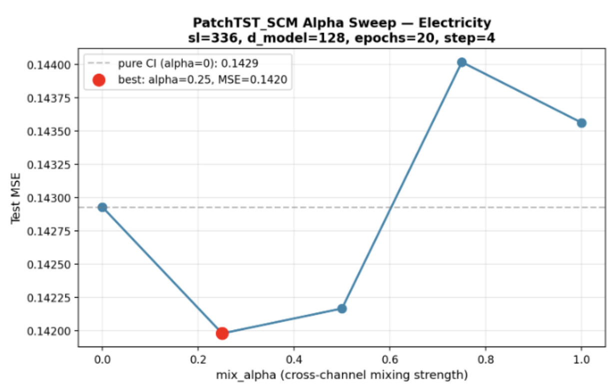

Figure 4: SCM interpolation sweep on Electricity. Optimal mixing at α=0.25; heavy mixing ($\alpha \geq$0.75) degrades below pure CI.